

2025 AMC 10B Problem 14 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 14



Problem:

Nine athletes, no two of whom are the same height, try out for the basketball team. One at a time, they draw a wristband at random, without replacement, from a bag containing 3 blue bands, 3 red bands, and 3 green bands. They are divided into a blue group, a red group, and a green group. The tallest member of each group is named the group captain. What is the probability that the group captains are the three tallest athletes?

Answer Choices:

A. $\frac{2}{9}$

B. $\frac{2}{7}$

C. $\frac{9}{28}$

D. $\frac{1}{3}$

E. $\frac{3}{8}$

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Solution:

Without loss of generality, suppose the 3 tallest athletes draw first. The condition is equivalent to saying they must all draw different colored wristbands. Regardless of which color band the first person draws, the second person has a $\frac{6}{8}$ chance to draw a different colored band on their draw. The third person then has a $\frac{3}{7}$ chance to draw a band different from the other two. Thus the final answer is

$$\frac{6}{8} \cdot \frac{3}{7} = \boxed{(C) \frac{9}{28}}$$

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